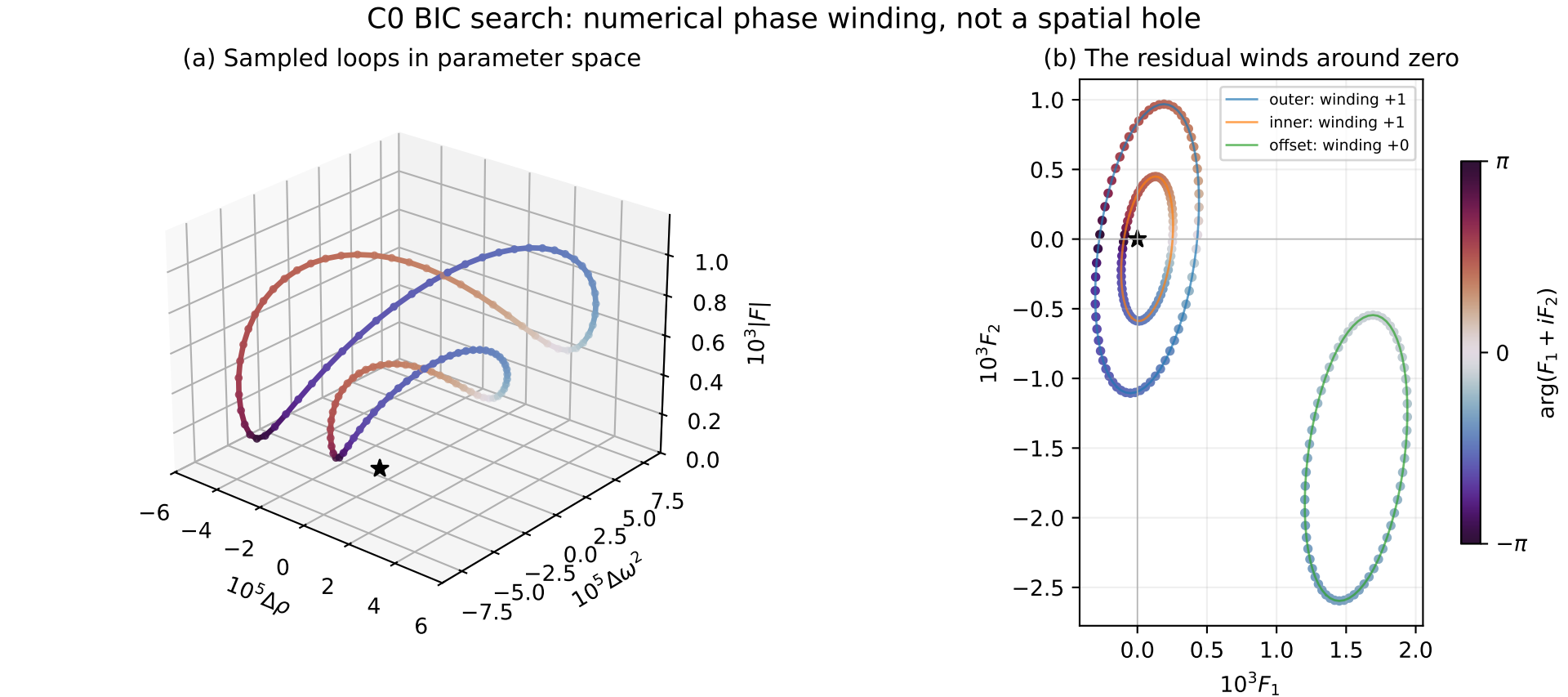

C0 BIC search: numerical phase winding, not a spatial hole
(a) Sampled loops in parameter space
(b) The residual winds around zero
outer: winding +1
inner: winding +1
offset: winding +0
$10^3|F|$
$10^5\Delta\rho$
$10^5\Delta\omega^2$
10^3F_1
10^3F_2
$\arg(F_1 + iF_2)$
Stored finite-R origin residual, R=44. Colored segments join samples only; no interpolated surface or flux units.